

ON THE ASYMPTOTIC DISTRIBUTION OF A NONPARAMETRIC ESTIMATOR OF A REGRESSION FUNCTION WHEN SAMPLE SIZE IS RANDOM

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Summary: In this paper the asymptotic distribution of the regression estimator of Ahmad and Lin (1976) has been studied for the case where the sample size $N(t)$ is a random variable depending on a positive parameter t . It has been shown that this estimator, when properly normalised, has an asymptotic normal distribution as t tends to infinity.

1. INTRODUCTION

It has been observed (see Srivastava 1973) that there are situations where it may not be possible to fix the sample size before the observations are taken. For example, in studying particular numerical characteristics of people arriving in a queue, say, at a hospital or clinic, or the number of immigrants that arrive in an area within a time period $(0, t]$, it is better to regard the sample size $N(t)$ as a random variable.

Let $(X_1, Y_1), (X_2, Y_2), \dots$ be a sequence of independent observations of a two-dimensional random variable (X, Y) having the joint probability

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density function $f(x,y)$. Let the marginal p.d.f. of X be denoted by $g(x)$ and the regression function of Y on X , $E[Y|X=x]$, be denoted by $r(x)$. Where the sample size n can be fixed, Ahmad and Lin (1976) have proposed an estimator $r_n(x)$ of $r(x)$ given by

$$(1) \quad r_n(x) = \frac{\sum_{j=1}^n \frac{1}{b_j} Y_j K\left(\frac{x-X_j}{b_j}\right)}{\sum_{j=1}^n \frac{1}{b_j} K\left(\frac{x-X_j}{b_j}\right)}$$

where K is a known p.d.f. and $\{b_j\}$ is a monotonic decreasing sequence of positive numbers converging to zero as j approaches infinity.

In this paper we have considered a regression estimator analogous to that in (1) when the sample size $N(t)$ is a random variable depending on a positive parameter t . We have assumed that for every $t > 0$, $N(t)$ is a positive integer-valued random variable such that $\frac{N(t)}{t}$ converges in probability to a positive constant π as t approaches infinity. In particular, if $\{N(t), t > 0\}$ is a homogeneous Poisson process with mean function $M(t) = \pi t$, $\pi > 0$, then this condition on $N(t)$ is satisfied (see theorem 4.1 of Thompson, Jr. (1981)).

On the basis of the sample $(X_1, Y_1), (X_2, Y_2), \dots, (X_{N(t)}, Y_{N(t)})$, which could be independent observations on, say, the age X and blood pressure Y measured on $N(t)$ patients arriving at a clinic within a time interval $(0, t]$, we have proposed an estimator $r_{N(t)}(x)$ of $r(x)$ given by

$$(2) \quad r_{N(t)}(x) = \frac{\sum_{j=1}^{N(t)} \frac{1}{b_j} Y_j K\left(\frac{x-X_j}{b_j}\right)}{\sum_{j=1}^{N(t)} \frac{1}{b_j} K\left(\frac{x-X_j}{b_j}\right)}.$$

We have shown that this estimator, when properly normalised, is asymptotically normally distributed as t tends to infinity.

Various authors have studied similar problems in density estimation when the sample size is a random variable. Srivastava (1973) looked at the problem of estimating a continuous p.d.f. using the kernel estimator of Parzen (1962) and Carroll (1976) studied this estimator and the recursive-type of Yamato (1971) for the case where the sample size is random. Davies and Wegman (1975) applied classical sequential statistical procedures to the estimation of density

estimators. Samanta and Mugisha (1981) studied various aspects in the estimation of a density function, including the asymptotic distributions of the estimators of a p.d.f. and the mode when the sample size is random. In this paper we have applied the technique of density estimation to the estimation of a regression function at a point.

In what follows the sequence $\{(X_i, Y_i)\}$ and the random variable $N(t)$ need not be independent. All integration, unless otherwise stated, is from $-\infty$ to ∞ .

2. ASYMPTOTIC NORMALITY OF THE ESTIMATOR $r_n(x)$

Let us define

$$\begin{aligned}(3) \quad w(x) &= \int y f(x, y) dy \\ r(x) &= \int y f(x, y) dy / \int f(x, y) dy = w(x) / g(x) \\ v(x) &= \int y^2 f(x, y) dy \\ w_n(x) &= \frac{1}{n} \sum_{j=1}^n \frac{1}{b_j} Y_j K\left(\frac{x - X_j}{b_j}\right) \\ g_n(x) &= \frac{1}{n} \sum_{j=1}^n \frac{1}{b_j} K\left(\frac{x - X_j}{b_j}\right) \\ q_n(x) &= w_n(x) - r(x) g_n(x).\end{aligned}$$

Then $r_n(x) - r(x) = q_n(x) / g_n(x)$.

In proving the asymptotic normality of $r_n(x)$ we shall need several lemmas. Lemma 1 is almost trivial and its proof is omitted. The proof in lemma 2 can be found in Samanta and Mugisha (1981) but is included here for completeness.

Lemma 1 Let $\{h_n\}$ be a sequence of functions converging to a function h as $n \rightarrow \infty$. Then $\frac{1}{n} \sum_{j=1}^n h_j(x)$ converges to $h(x)$ as $n \rightarrow \infty$.

Lemma 2 Let $\{s_{jn}, j=1, \dots, n; n=1, 2, \dots\}$ be a uniformly bounded double array of numbers such that $\frac{1}{n} \sum_{j=1}^n s_{jn}$ converges to s as $n \rightarrow \infty$. If h_j is a sequence of functions converging to a function h at a point x as $j \rightarrow \infty$, then $\frac{1}{n} \sum_{j=1}^n s_{jn} h_j(x)$ converges to $sh(x)$ as $n \rightarrow \infty$.

Proof We have

$$\left| \frac{1}{n} \sum_{j=1}^n s_{jn} h_j(x) - sh(x) \right| \leq \frac{1}{n} \sum_{j=1}^n |s_{jn}| |h_j(x) - h(x)| + |h(x)| \left| \frac{1}{n} \sum_{j=1}^n s_{jn} - s \right|$$

and the proof now follows from lemma 1.

Lemma 3 Let the density K be a bounded function satisfying the conditions $\sup_{-\infty < u < \infty} |uK(u)| < \infty$, $\int tK(t)dt = 0$ and $\int t^2 K(t)dt < \infty$, and let the

sequence $\{b_j\}$ satisfy the conditions $\lim_{n \rightarrow \infty} nb_n^5 = 0$ and $\frac{1}{n} \sum_{j=1}^n b_j^2 \leq Cb_n^2$ for $C > 0$. If w' , w'' , g' , and g'' exist and are bounded, then

$$(nb_n)^{\frac{1}{2}} q_n(x) = (nb_n)^{\frac{1}{2}} (q_n(x) - Eq_n(x)) + p_n(x)$$

where $p_n(x)$ converges to zero as $n \rightarrow \infty$.

Proof We have

$$p_n(x) = (nb_n)^{\frac{1}{2}} (Ew_n(x) - r(x)Eq_n(x)).$$

Now

$$Ew_n(x) = \frac{1}{n} \sum_{j=1}^n \frac{1}{b_j} E(YK(\frac{x-X}{b_j})),$$

and for any fixed j ,

$$\begin{aligned}
 E\left(\frac{1}{b_j} Y K\left(\frac{x-X}{b_j}\right)\right) &= \iint \frac{1}{b_j} y K\left(\frac{x-u}{b_j}\right) f(u, y) dy du \\
 &= \int \frac{1}{b_j} K\left(\frac{x-u}{b_j}\right) w(u) du \\
 &= \int w(x-b_j t) K(t) dt \\
 &= w(x) + \frac{1}{2} b_j^2 \int w''(x-\theta' b_j t) t^2 K(t) dt, \text{ where } 0 < \theta < 1.
 \end{aligned}$$

Similarly,

$$\begin{aligned}
 r(x) E\left(\frac{1}{b_j} K\left(\frac{x-X}{b_j}\right)\right) &= r(x) \left[g(x) + \frac{1}{2} b_j^2 \int g''(x-\theta' b_j t) t^2 K(t) dt \right] \\
 &= w(x) + \frac{1}{2} r(x) b_j^2 \int g''(x-\theta' b_j t) t^2 K(t) dt,
 \end{aligned}$$

where $0 < \theta' < 1$. Hence, letting C be a generic positive constant, we obtain

$$|E w_n(x) - r(x) E g_n(x)| \leq C \left(\frac{1}{n} \sum_{j=1}^n b_j^2 \right) \leq C b_n^2$$

and therefore,

$$(n b_n)^{\frac{1}{2}} (E w_n(x) - r(x) E g_n(x)) = O(n b_n^{\frac{5}{2}})^{\frac{1}{2}} = o(1).$$

Lemma 4 Let the density K satisfy the conditions $\sup_{-\infty < u < \infty} |K(u)| < \infty$,

$$\sup_{-\infty < u < \infty} |u K(u)| < \infty, \quad \lim_{|u| \rightarrow \infty} |u K(u)| = 0, \quad \int t K(t) dt = 0, \quad \int t^2 K(t) dt < \infty,$$

and let the sequence $\{b_j\}$ satisfy the conditions $\lim_{n \rightarrow \infty} n b_n^p = \infty$ for $p < 5$,

$$\lim_{n \rightarrow \infty} n b_n^5 = 0, \quad \frac{1}{n} \sum_{j=1}^n b_j^2 \leq C b_n^2, \quad C > 0, \quad \text{and} \quad \lim_{n \rightarrow \infty} \left(\frac{1}{n} \sum_{j=1}^n \frac{b_n}{b_j} \right) = a. \quad \text{If } g', w',$$

$v', g'',$ and w'' exist and are bounded, and if $E|Y|^{\frac{5}{2}+\eta}$ is finite for

arbitrarily small η , then $(n b_n)^{\frac{1}{2}} q_n(x)$ converges in distribution to a

normal random variable with mean zero and variance

$$\sigma^2(x) = a \operatorname{Var}(Y|X=x) g(x) \int K^2(t) dt \text{ as } n \rightarrow \infty.$$

Proof We have

$$\text{Var}(q_n(x)) = \text{Var}(w_n(x)) - 2r(x)\text{Cov}(w_n(x), g_n(x)) + r^2(x)\text{Var}(g_n(x)).$$

Now

$$(4) \quad nb_n \text{Var}(w_n(x)) = \frac{1}{n} \sum_{j=1}^n \frac{b_n}{b_j} E \left(\frac{1}{b_j} Y^2 K^2 \left(\frac{x-X}{b_j} \right) \right) - \frac{b_n}{n} \sum_{j=1}^n \left(E \frac{1}{b_j} Y K \left(\frac{x-X}{b_j} \right) \right)^2.$$

But

$$\begin{aligned} |E(\frac{1}{b_j} Y^2 K^2(\frac{x-X}{b_j})) - v(x) \int K^2(t) dt| &= |\int \{v(x-b_j t) - v(x)\} K^2(t) dt| \\ &\leq \sup_{-\infty < u < \infty} |v'(u)| b_j \int |t| K^2(t) dt, \end{aligned}$$

which converges to zero as $j \rightarrow \infty$. It is not hard to show that the second term in the r.h.s. of (4) converges to zero as $n \rightarrow \infty$. We now conclude from lemma 2 that $nb_n \text{Var}(w_n(x))$ converges to $a v(x) \int K^2(t) dt$ as $n \rightarrow \infty$. In a similar manner we get

$$\lim_{n \rightarrow \infty} (nb_n \text{Cov}(w_n(x), g_n(x))) = a w(x) \int K^2(t) dt$$

and

$$\lim_{n \rightarrow \infty} (nb_n \text{Var}(g_n(x))) = a g(x) \int K^2(t) dt.$$

Therefore,

$$\lim_{n \rightarrow \infty} (nb_n \text{Var}(q_n(x))) = a \text{Var}(Y|X=x) g(x) \int K^2(t) dt.$$

From lemma 3, it will now suffice to show, using the Liapunov's form of the central limit theorem, that for some $\delta > 0$,

$$(n^{-1} b_n)^{1+\frac{\delta}{2}} \left[\sum_{j=1}^n E \left| \frac{1}{b_j} (Y-r(x)) K \left(\frac{x-X}{b_j} \right) - E \left\{ \frac{1}{b_j} (Y-r(x)) K \left(\frac{x-X}{b_j} \right) \right\} \right|^{2+\delta} \right] / (nb_n \text{Var}(q_n(x)))^{1+\frac{\delta}{2}}$$

converges to zero as $n \rightarrow \infty$. By choosing $\delta = \frac{1}{2}$ and applying the C_r inequality of Loeve (1963) successively, it will suffice to show that

$$\lim_{n \rightarrow \infty} \left[(n^{-1}b_n)^{\frac{5}{4}} \sum_{j=1}^n E \left| \frac{1}{b_j} Y K \left(\frac{X-X}{b_j} \right) \right|^{\frac{5}{2}} \right] = 0.$$

Using Holder's inequality, we have for any $\varepsilon > 0$, and any j ,

$$\begin{aligned} & E \left\{ \left| \frac{1}{b_j} Y K \left(\frac{X-X}{b_j} \right) \right|^{\frac{5}{2}} \right\} \\ & \leq \left(\frac{1}{b_j} \right)^{\frac{5}{2}} [E|Y|^{\frac{5}{2}(1+\varepsilon)}]^{\frac{1}{1+\varepsilon}} [E|K(\frac{X-X}{b_j})|^{\frac{5}{2}(\frac{1+\varepsilon}{\varepsilon})}]^{\frac{\varepsilon}{1+\varepsilon}} \\ & = \left(\frac{1}{b_j} \right)^{\frac{5}{2} - \frac{\varepsilon}{1+\varepsilon}} [E|Y|^{\frac{5}{2}(1+\varepsilon)}]^{\frac{1}{1+\varepsilon}} \left[\frac{1}{b_j} E|K(\frac{X-X}{b_j})|^{\frac{5}{2}(\frac{1+\varepsilon}{\varepsilon})} \right]^{\frac{\varepsilon}{1+\varepsilon}}. \end{aligned}$$

From the fact that $\sup_{-\infty < u < \infty} |K(u)| < \infty$, $\lim_{|u| \rightarrow \infty} |uK(u)| = 0$, and $\int |K(t)| dt < \infty$,

we conclude from theorem 1A of Parzen (1962) that as $j \rightarrow \infty$, $\frac{1}{b_j} E|K(\frac{X-X}{b_j})|^{\frac{5}{2}(\frac{1+\varepsilon}{\varepsilon})}$ converges to $g(x) \int |K(t)|^{\frac{5}{2}(\frac{1+\varepsilon}{\varepsilon})} dt$ and hence the sequence $\left\{ \frac{1}{b_j} E|K(\frac{X-X}{b_j})|^{\frac{5}{2}(\frac{1+\varepsilon}{\varepsilon})} \right\}$ is bounded.

Therefore, using the monotonicity of the sequence $\{b_j\}$ we obtain

$$\begin{aligned} (n^{-1}b_n)^{\frac{5}{4}} \sum_{j=1}^n E \left| \frac{1}{b_j} Y K \left(\frac{X-X}{b_j} \right) \right|^{\frac{5}{2}} & \leq C n (n^{-1}b_n)^{\frac{5}{4}} \left(\frac{1}{b_n} \right)^{\frac{5}{2} - \frac{\varepsilon}{1+\varepsilon}} \\ & = C (nb_n)^{5 - \frac{4\varepsilon}{1+\varepsilon} - \frac{1}{4}} \end{aligned}$$

which converges to zero as n tends to infinity. This completes the proof.

The following lemma is due to Davies (1973) and its proof is omitted.

Lemma 5. If the density $K(u)$ satisfies the conditions $\sup_{-\infty < u < \infty} |K(u)| < \infty$ and

$\lim_{|u| \rightarrow \infty} |uK(u)| = 0$, and if the sequence $\{b_n\}$ is chosen such that

$$\sum_{n=1}^{\infty} (n^2 b_n)^{-1} < \infty \quad \text{and} \quad \lim_{n \rightarrow \infty} \left[\frac{1}{n} \sum_{j=1}^n \frac{b_n}{b_j} \right] = a, \quad 0 < a < 1, \quad \text{then, as } n \rightarrow \infty,$$

$g_n(x) \rightarrow g(x)$ with probability one if x is a continuity point of g .

3. ASYMPTOTIC NORMALITY OF THE ESTIMATOR $r_{N(t)}(x)$

In what follows we shall assume that for every $t > 0$, $N(t)$ is an integer-valued random variable such that $\frac{N(t)}{t}$ converges in probability to a positive constant π as t tends to infinity.

On the basis of the sample $(X_1, Y_1), (X_2, Y_2), \dots, (X_{N(t)}, Y_{N(t)})$, we define

$$(5) \quad w_{N(t)}(x) = \frac{1}{N(t)} \sum_{j=1}^{N(t)} \frac{1}{b_j} Y_j K\left(\frac{x - X_j}{b_j}\right)$$

$$g_{N(t)}(x) = \frac{1}{N(t)} \sum_{j=1}^{N(t)} \frac{1}{b_j} K\left(\frac{x - X_j}{b_j}\right)$$

and

$$q_{N(t)}(x) = w_{N(t)}(x) - r(x)g_{N(t)}(x).$$

Then $r_{N(t)}(x) - r(x) = q_{N(t)}/g_{N(t)}$.

Lemma 6 Suppose the density function K and the sequence $\{b_j\}$ satisfy the conditions in lemma 5. Then $g_{N(t)}(x)$ converges in probability to $g(x)$ as $t \rightarrow \infty$ if x is a continuity point of g .

Proof Let ε and ε' be arbitrarily small positive numbers. By lemma 5 we can find an integer $n_0 = n_0(\varepsilon, \varepsilon')$ such that

$$P\left[|g_n(x) - g(x)| < \frac{\varepsilon}{2} \text{ for all } n \text{ greater than } n_0\right] \geq 1 - \varepsilon'$$

and

$$P\{|g_{m+n}(x) - g(x)| < \frac{\varepsilon}{2} \text{ for all } m \text{ whenever } n > n_0\} \geq 1 - \varepsilon'.$$

We can also find a $t_0 = t_0(\varepsilon, \varepsilon')$ such that for $t > t_0$,

$$P\{|N(t) - \pi t| < \varepsilon t\} \geq 1 - \varepsilon'.$$

Let $N' = [t(\pi - \varepsilon)]$ and $N'' = [t(\pi + \varepsilon)] + 1$, where $[z]$ denotes the integer part of z . We now choose $t > t_0$ such that $N' > n_0$. Then we get

$$\begin{aligned} & P\{|g_{N(t)}(x) - g(x)| < \varepsilon\} \\ & \geq P\{|g_{N(t)}(x) - g(x)| < \varepsilon; |N(t) - \pi t| < \varepsilon t; |g_{N(t)}(x) - g_n(x)| < \frac{\varepsilon}{2} \text{ for} \\ & \quad \text{some integer } n \text{ lying between } N' \text{ and } N''\} \\ & \geq P\{|g_n(x) - g(x)| < \frac{\varepsilon}{2}; |N(t) - \pi t| < \varepsilon t; |g_n(x) - g_m(x)| < \frac{\varepsilon}{2} \text{ for all} \\ & \quad \text{integers } m \text{ and } n \text{ lying between } N' \text{ and } N''\} \\ & \geq P\{|g_n(x) - g(x)| < \frac{\varepsilon}{2} \text{ for all } n \text{ lying between } N' \text{ and } N''\} - P\{|N(t) - \pi t| \geq \varepsilon t\} \\ & \quad - P\{|g_n(x) - g_m(x)| \geq \frac{\varepsilon}{2} \text{ for all } m \text{ and } n \text{ lying between } N' \text{ and } N''\} \\ & \geq 1 - 3\varepsilon'. \end{aligned}$$

We are now in a position to prove the following theorem.

Theorem Let the density K and the sequence $\{b_j\}$ satisfy the conditions

$$\sup_{-\infty < u < \infty} |K(u)| < \infty, \quad \lim_{|u| \rightarrow \infty} |uK(u)| = 0, \quad \sup_{-\infty < u < \infty} |uK(u)| < \infty, \quad \int tK(t)dt = 0,$$

$\int t^2 K(t)dt < \infty$ and $\{b_j = j^{-\delta}\}$ where $\frac{1}{5} < \delta < 1$. Suppose $g', w', v', g'',$ and w'' exist and are bounded and, further, suppose $E|Y|^{(\frac{5}{2})+\eta} < \infty$

for arbitrarily small η . If, for every $t > 0$, $N(t)$ is an integer-valued random variable such that $\frac{N(t)}{t} \rightarrow \pi > 0$ in probability as $t \rightarrow \infty$, then

$(N(t)b_{N(t)})^{\frac{1}{2}} (r_{N(t)}(x) - r(x))$ converges in distribution to a normal random

variable with mean zero and variance $(1+\delta)^{-1} \text{Var}(Y|X=x) \int k^2(u) du / g(x)$ as $t \rightarrow \infty$.

Proof The proof resembles that of theorem 1 in Renyi (1957). Following from lemma 6, it will suffice to show that as $t \rightarrow \infty$, $N(t)b_{N(t)}^{\frac{1}{2}}q_{N(t)}(x)$ converges in distribution to a normal random variable with mean zero and variance $(1+\delta)^{-1} \text{Var}(Y|X=x) g(x) \int k^2(u) du$.

Now, let $\varepsilon > 0$ be arbitrarily small. Then there exists a $t_0 = t_0(\varepsilon)$ such that for $t > t_0$,

$$P[|N(t) - \pi t| \geq \pi t \varepsilon] < \varepsilon.$$

We define

$$(6) \quad N_1 = [\pi t(1-\varepsilon)] \quad \text{and} \quad N_2 = [\pi t(1+\varepsilon)]$$

where $[z]$ is the largest integer contained in z . For any $0 < \varepsilon < \frac{1}{2}$ and $t > \frac{1}{\pi \varepsilon}$ the following inequalities are satisfied:

$$(7) \quad \frac{N_2}{N_1} < \frac{1+\varepsilon}{1-2\varepsilon} \quad \text{and} \quad \frac{N_2 - N_1}{N_1} < \frac{3\varepsilon}{1-2\varepsilon}.$$

Define

$$(8) \quad Z_j(x) = \frac{1}{b_j} (Y_j - r(x)) K\left(\frac{x - X_j}{b_j}\right).$$

Then

$$(9) \quad (N(t)b_{N(t)})^{\frac{1}{2}}q_{N(t)}(x) = ((N(t))^{-1}b_{N(t)})^{\frac{1}{2}} \sum_{j=1}^{N(t)} Z_j(x).$$

For $y > 0$ we have

$$\begin{aligned} & P[(N(t)b_{N(t)})^{\frac{1}{2}}q_{N(t)}(x) < y] \\ &= \sum_{|n-\pi t| < \pi t \varepsilon} P[(nb_n)^{\frac{1}{2}}q_n(x) < y; N(t)=n] + \sum_{|n-\pi t| \geq \pi t \varepsilon} P[(nb_n)^{\frac{1}{2}}q_n(x) < y; N(t)=n] \end{aligned}$$

$$\leq \sum_{|n-\pi t| < \pi t \varepsilon} P[(nb_n)^{\frac{1}{2}} q_n(x) < y; N(t) = n] + P[|N(t) - \pi t| \geq \pi t \varepsilon]$$

$$\leq \sum_{|n-\pi t| < \pi t \varepsilon} P[(nb_n)^{\frac{1}{2}} q_n(x) < y; N(t) = n] + \varepsilon.$$

Hence

$$(10) \quad |P[(N(t)b_{N(t)})^{\frac{1}{2}} q_{N(t)}(x) < y - \sum_{|n-\pi t| < \pi t \varepsilon} P[(nb_n)^{\frac{1}{2}} q_n(x) < y; N(t) = n]| \leq \varepsilon.$$

For any fixed n in the half open interval $(N_1, N_2]$

$$(11) \quad P[(nb_n)^{\frac{1}{2}} q_n(x) < y; N(t) = n]$$

$$= P\left[\sum_{j=1}^{N_1} Z_j(x) + \sum_{j=N_1+1}^n (Z_j(x) - EZ_j(x)) + \sum_{j=N_1+1}^n EZ_j(x) < y(n^{-1}b_n)^{-\frac{1}{2}}; N(t)=n\right]$$

$$(12) \leq P\left[(N_1^{-1}b_{N_1})^{\frac{1}{2}} \sum_{j=1}^{N_1} Z_j(x) < y\left(\frac{N_2}{N_1}\right)^{\frac{1}{2}} (b_{N_1}/b_{N_2})^{\frac{1}{2}} + M(x) + (N_1^{-1}b_{N_1})^{\frac{1}{2}} \left| \sum_{j=N_1+1}^n EZ_j(x) \right|; N(t) = n\right],$$

$$\text{where } M(x) = \max_{N_1 < n \leq N_2} |(N_1^{-1}b_{N_1})^{\frac{1}{2}} \sum_{j=N_1+1}^n (Z_j(x) - EZ_j(x))|.$$

By Kolmogorov's inequality,

$$P[M(x) > \varepsilon^{1/3}] \leq \sum_{j=N_1+1}^{N_2} \text{Var}(Z_j(x)) / (\varepsilon^{2/3} N_1 b_{N_1}^{-1})$$

$$\leq C(N_2 - N_1) b_{N_1} \varepsilon^{-2/3} (N_1 b_{N_2})^{-1}$$

$$= C \varepsilon^{-2/3} \left(\frac{N_2 - N_1}{N_1}\right) \left(\frac{N_2}{N_1}\right)^\delta$$

$$\leq \frac{C \varepsilon^{1/3} (1+\varepsilon)^\delta}{(1-2\varepsilon)^{1+\delta}}.$$

In like manner we can show that for any n in the interval $(N_1, N_2]$

$$\begin{aligned}
 (N_1^{-1} b_{N_1})^{\frac{1}{2}} \left| \sum_{j=N_1+1}^n E Z_j(x) \right| &\leq C(N_1^{-1} b_{N_1})^{\frac{1}{2}} (N_2 - N_1) b_{N_1}^2 \\
 &= C \left(\frac{N_2 - N_1}{N_1} \right) (N_1 b_{N_1}^5)^{\frac{1}{2}} \leq \frac{C\epsilon}{1-2\epsilon}.
 \end{aligned}$$

We now conclude that the r.h.s. in (12) is less than or equal to

$$P[(N_1 b_{N_1})^{\frac{1}{2}} q_{N_1}(x) < y \left(\frac{N_2}{N_1} \right)^{\frac{1}{2}} (b_{N_1}/b_{N_2})^{\frac{1}{2}} + M(x) + \frac{C\epsilon}{1-2\epsilon}; N(t) = n]$$

and therefore,

$$\begin{aligned}
 (13) \quad &\sum_{|n-\pi t| < \pi t \epsilon} P[(nb_n)^{\frac{1}{2}} q_n(x) < y; N(t) = n] \\
 &\leq P[(N_1 b_{N_1})^{\frac{1}{2}} q_{N_1}(x) < y \left(\frac{1+\epsilon}{1-2\epsilon} \right)^{\frac{1+\delta}{2}} + \epsilon^{1/3} + \frac{C\epsilon}{1-2\epsilon}; |N(t) - \pi t| < \pi t \epsilon] + P[M(x) \geq \epsilon^{1/3}] \\
 &\leq P[(N_1 b_{N_1})^{\frac{1}{2}} q_{N_1}(x) < y \left(\frac{1+\epsilon}{1-2\epsilon} \right)^{\frac{1+\delta}{2}} + \epsilon^{1/3} + \frac{C\epsilon}{1-2\epsilon}] + \frac{C\epsilon^{1/3}}{1-2\epsilon} \left(\frac{1+\epsilon}{1-2\epsilon} \right)^{\delta}.
 \end{aligned}$$

From (11) we can get in a similar manner

$$\begin{aligned}
 (14) \quad &P[(nb_n)^{\frac{1}{2}} q_n(x) < y; N(t) = n] \\
 &\geq P[(N_1 b_{N_1})^{\frac{1}{2}} q_{N_1}(x) < y - \frac{C\epsilon}{1-2\epsilon} - M(x); |N(t) - \pi t| < \pi t \epsilon] \\
 &\geq P[(N_1 b_{N_1})^{\frac{1}{2}} q_{N_1}(x) < y - \frac{C\epsilon}{1-2\epsilon} - \epsilon^{1/3}; M(x) < \epsilon^{1/3}; |N(t) - \pi t| < \pi t \epsilon] \\
 &\geq P[(N_1 b_{N_1})^{\frac{1}{2}} q_{N_1}(x) < y - \frac{C\epsilon}{1-2\epsilon} - \epsilon^{1/3}] - P[M(x) \geq \epsilon^{1/3}] - P[|N(t) - \pi t| \geq \pi t \epsilon] \\
 &\geq P[(N_1 b_{N_1})^{\frac{1}{2}} q_{N_1}(x) < y - \frac{C\epsilon}{1-2\epsilon} - \epsilon^{1/3}] - \left(\frac{C\epsilon^{1/3}}{1-2\epsilon} \right) \left(\frac{1+\epsilon}{1-2\epsilon} \right)^{\delta} - \epsilon.
 \end{aligned}$$

From (10), (13), and (14), we conclude that for $t > t_0$

$$P[(N_1 b_{N_1})^{\frac{1}{2}} q_{N_1}(x) < y - \frac{C\epsilon}{1-2\epsilon} - \epsilon^{1/3}] - \frac{C\epsilon^{1/3} (1+\epsilon)^{\delta}}{(1-2\epsilon)^{1+\delta}} - 2\epsilon \leq$$

$$P[(N(t)b_{N(t)})^{1/2}q_{N(t)}(x) < y] \leq P[(N_1b_{N_1})^{1/2}q_{N_1}(x) < y(\frac{1+\epsilon}{1-2\epsilon})^{1/2} + \epsilon^{1/3} + \frac{C\epsilon}{1-2\epsilon}] \\ + \frac{C\epsilon^{1/3}(1+\epsilon)^{\delta}}{(1-2\epsilon)^{1+\delta}} + \epsilon.$$

Similar statements hold for $y < 0$. We now apply lemma 4 and use the continuity of the distribution function of the normal random variable to complete the proof.

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